

TOPIC 3.2

How is energy transformed?

Key Concepts

- Energy is transformed in chemical reactions.
- Energy is transformed in nuclear reactions.
- Energy is transformed when light energy interacts with matter.

Curricular Competencies

- Collaboratively and individually plan, select, and use appropriate investigative methods.
- Evaluate your methods and experimental conditions.
- Transfer and apply learning to new situations.
- Formulate physical or mental theoretical models to describe a phenomenon.

About 66 million years ago, a meteor hurtled through space, plunged through Earth's atmosphere, and struck what is now the Yucatan Peninsula in Mexico. The impact sent vast amounts of dust and debris into the atmosphere as the mechanical kinetic energy of the meteorite was transferred to Earth. About 150 000 000 km away in the Sun, nuclear energy continued to be transformed into solar energy as it had been for billions of years. Now, however, the particles in Earth's atmosphere kept some of that energy from reaching the surface. Many scientists believe that this contributed to the extinction of the dinosaurs and many other species at the time. In doing so, it may have given a competitive edge to another group of animals—the mammals.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Inferring** Why could a reduction of solar energy at Earth's surface result in the extinction of so many species 66 million years ago?
2. **Modelling** Observe as your teacher adds water to a clear basin on an overhead projector. With the room dark and projector on, watch as coffee creamer is added to the water a few times. How does this system model the effects of dust on the atmosphere? How might these effects influence energy transformation?
3. **Applying First Peoples Perspectives** Find examples of traditional First Peoples' food processing techniques that rely on the transformation of energy to preserve food for the winter.



Key Terms

There are six key terms that are highlighted in bold type in this Topic:

- radiation
- nuclear decay
- nuclear fission
- chain reaction
- nuclear fusion
- absorption

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

Energy is transformed in chemical reactions.

Activity

That's How It Glows



Materials

2 glow sticks

2 beakers

2 pairs of tongs

2 thermometers

ice water

hot water

Design a procedure that uses the above materials to determine how energy transfer and/or transformation can be affected by adding or removing thermal energy from a system. Get your teacher's approval to carry it out.

Connect to Unit 2, Topic 2.3

How is chemical potential energy transformed to run a car or contract a muscle, as in **Figure 3.12**? How can other forms of energy be transformed into chemical potential energy?



Figure 3.12 Chemical potential energy is transformed to contract a muscle **A** or run a race car **B**.

Energy Transformation in Chemical Reactions

Chemical reactions can transform some of the energy stored in chemical bonds into other forms of energy. The materials involved in a chemical reaction can be thought of as a system. In all chemical reactions, energy is released to the surroundings or absorbed from the surroundings. This energy tends to be thermal energy, but other forms, such as sound and light, can be released as well (**Figure 3.13**).



Figure 3.13 **A** Luminol undergoes a chemical reaction at crime scenes, releasing light energy that makes blood visible to investigators. **B** Fireworks undergo chemical reactions to release both light and sound.

The amount of energy transformed in a chemical reaction is determined by the nature of the chemical bonds in the reactants and products. The amount of energy stored in the bonds of some compounds is more, or less, than in others.

- Some compounds would release little energy if formed from elements. Their bonds give these compounds high chemical potential energy. The compounds have the *potential* of releasing more energy by undergoing chemical reactions and forming different compounds.
- Some compounds would release a lot of energy if formed from elements. Their bonds give these compounds low chemical potential energy. Their *potential* for releasing more energy by undergoing chemical reactions and forming different compounds is low.

Whether a chemical reaction releases or absorbs energy is determined as follows.

- If the reactants are higher in chemical potential energy than the products, energy is released by the system during the reaction. The reaction is *exothermic*.
- If the reactants are lower in chemical potential energy than the products, energy is absorbed from the surroundings during the reaction. The reaction is *endothermic*.

Figure 3.14 reviews how this works with the combustion of methane—a common reaction in gas stoves, furnaces, and Bunsen burners.

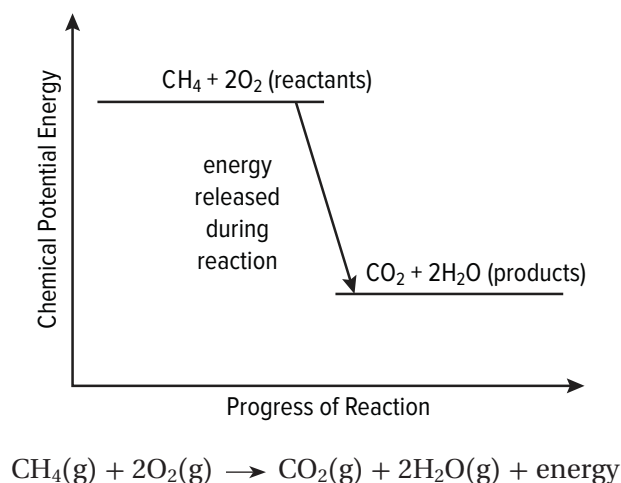


Figure 3.14 When compounds that are higher in chemical potential energy react to form compounds that are lower in chemical potential energy, energy is released in the reaction.

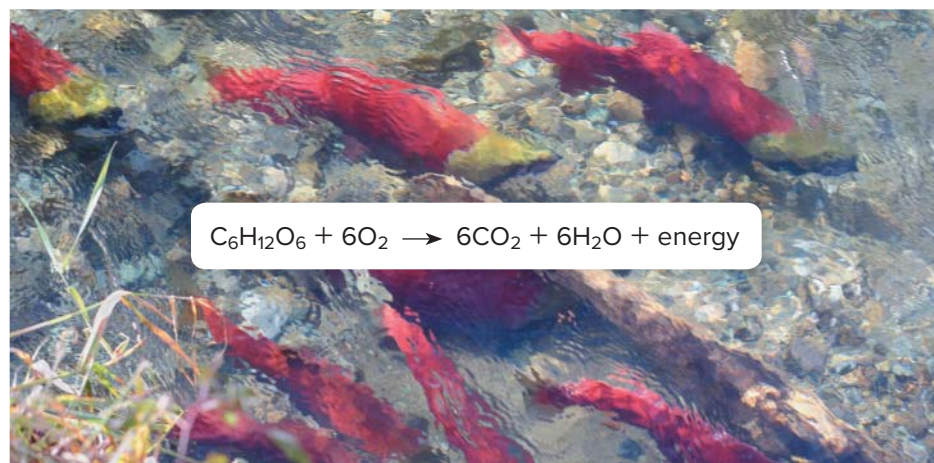
The reactants, methane and oxygen, are higher in chemical potential energy; the products, carbon dioxide and water, are lower in chemical potential energy. Thus, when methane reacts with oxygen to form carbon dioxide and water, a large amount of energy is released.

Chemical Reactions in Animals and Plants

Cellular Respiration

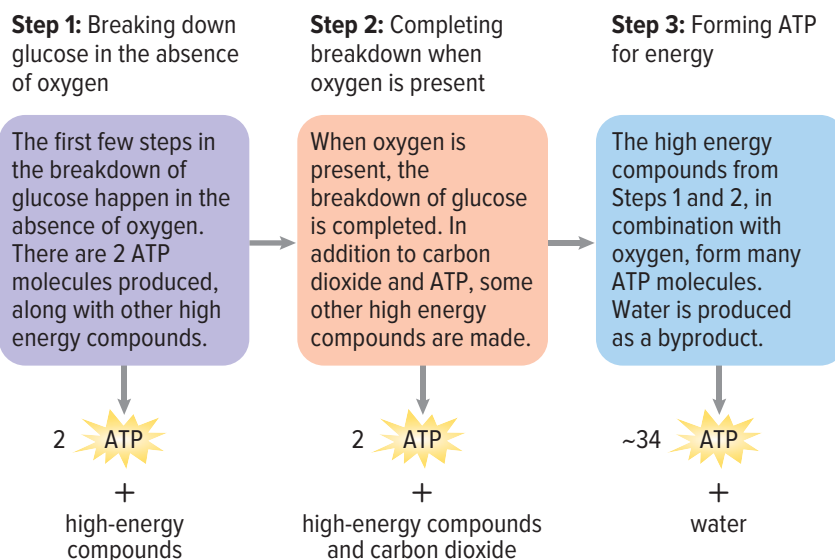
All living things, including us, transform chemical potential energy to carry out life processes. This generally occurs through *cellular respiration*—a series of chemical reactions in which glucose, $C_6H_{12}O_6$, and oxygen react to form carbon dioxide and water (Figure 3.15). The amount of energy released is similar to the amount released when methane reacts with oxygen. Your body does not combust or burn, however, because the overall reaction takes place in small, controlled steps.

Figure 3.15 This equation for cellular respiration summarizes a series of reactions.



During cellular respiration (Figure 3.16), some of the chemical potential energy in the glucose bonds is used to make bonds in a molecule called adenosine triphosphate (ATP). ATP is then used in nearly all reactions that require energy in living tissues. For instance, some of the energy is transformed into mechanical kinetic energy when muscles contract, some is transformed into more chemical potential energy when new molecules are created, and some is transformed into electrical potential energy, which can then be used to send signals along nerve cells.

Figure 3.16 Living things get energy through a series of chemical reactions in cellular respiration. In total, up to 38 ATP molecules are produced in three main steps. **Applying:** How is energy lost to the surroundings during cellular respiration?



Photosynthesis

If glucose is constantly undergoing chemical reactions in living things, where does this glucose come from? The answer is from other chemical reactions. In *photosynthesis*, carbon dioxide and water combine, with the addition of light energy, in a series of reactions to produce glucose and oxygen (**Figure 3.17**). Photosynthesis occurs mainly in plants, although it also occurs in algae and some other microscopic organisms.

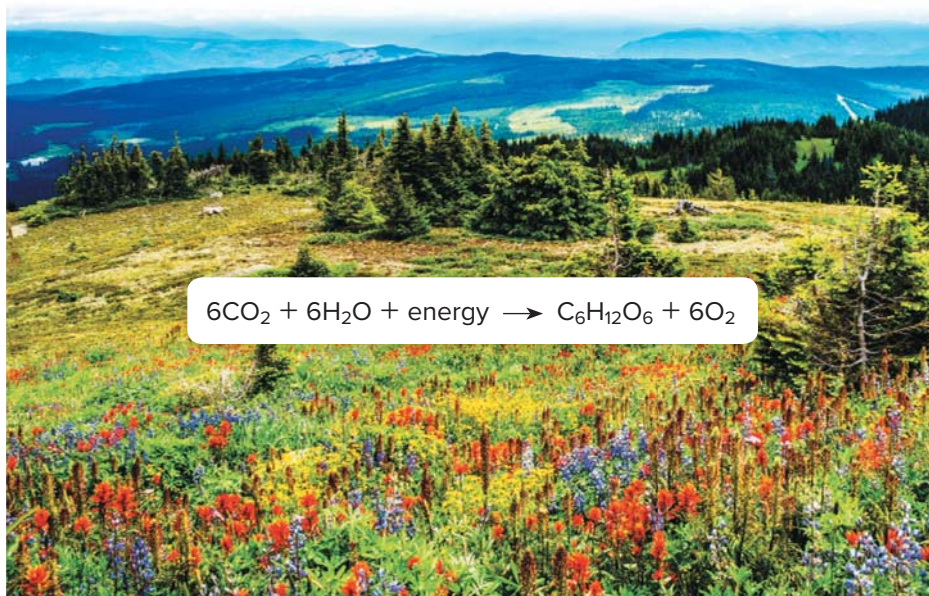


Figure 3.17 This equation for photosynthesis summarizes a series of reactions.

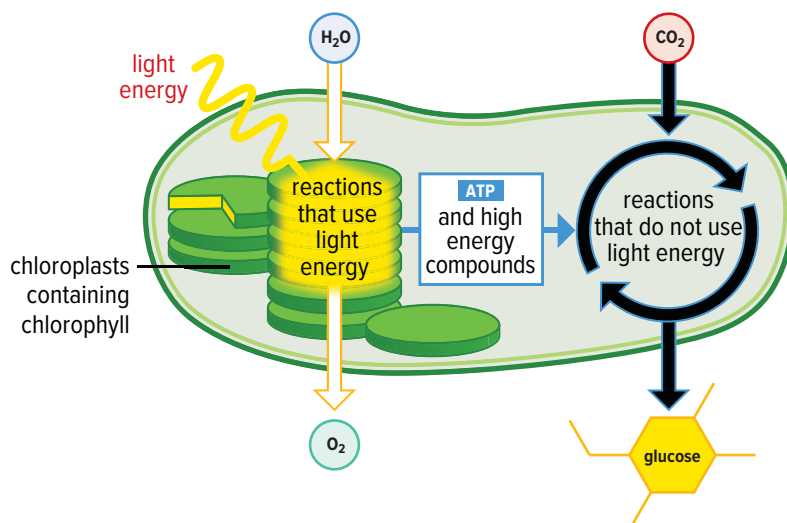
In plants, photosynthesis occurs in structures called chloroplasts through a series of reactions (**Figure 3.18**). In these reactions, light energy is transformed into the chemical potential energy stored in the bonds of glucose molecules.

Figure 3.18 Plants transform light energy to chemical potential energy through photosynthesis. Two sets of reactions occur—ones that need light energy and ones that don't.

A green molecule in the chloroplasts, called chlorophyll, absorbs incoming light energy.

The chlorophyll molecules become excited and pass the energy to special compounds, which use it to make ATP and other high energy compounds. Water is also used in these reactions, and oxygen is produced.

Another set of reactions can occur without light. These reactions use carbon dioxide, ATP, and the other high energy compounds to make glucose.

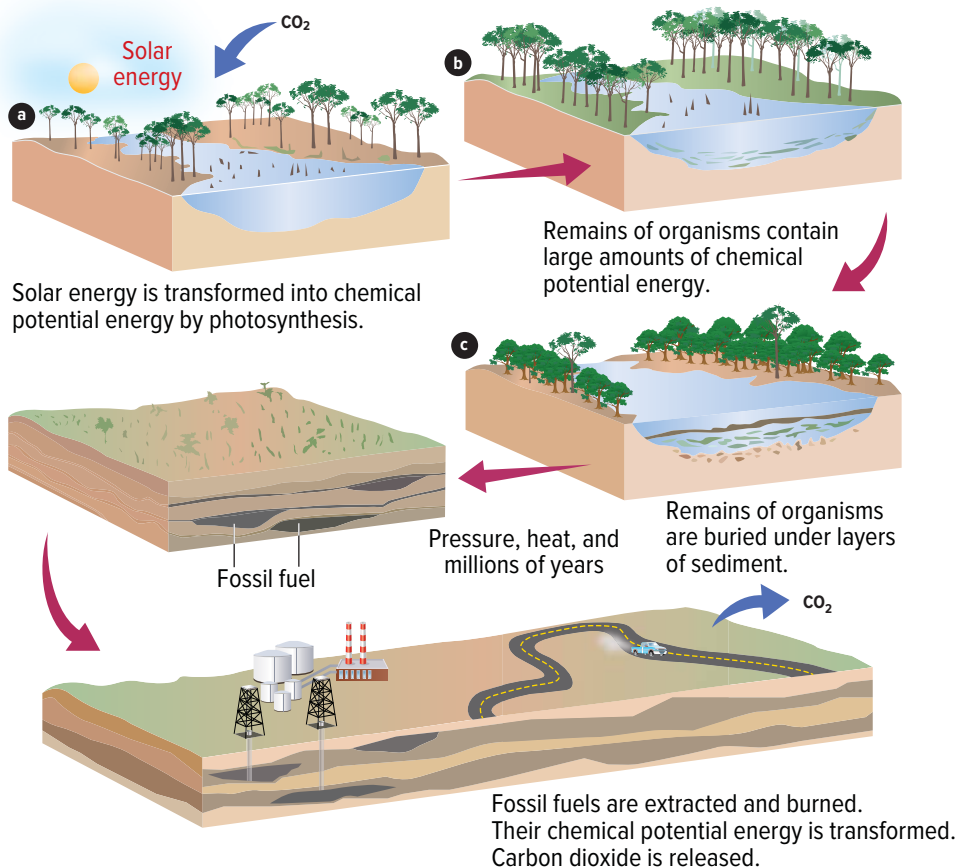


Energy Transformation and Fuels

Fossil Fuels

Fossil fuels contain large amounts of chemical potential energy that was transformed from solar energy by ancient plants. **Figure 3.19** shows how they became buried under the right conditions to form these fuels. Certain bacterial decay processes and long periods of time under high pressure eventually produced deposits of coal, oil, and natural gas.

Figure 3.19 Solar energy is transformed into chemical potential energy in living things, which ultimately provides the energy in fossil fuels. **Analyzing:** How is the transformation of chemical potential energy from fossil fuels by humans helpful? How is it harmful?



Fossil Fuel Combustion

It took millions of years for fossil fuels to accumulate on Earth. Today, we are extracting these energy resources very rapidly. As we burn and process fossil fuels, tonnes of carbon dioxide are released into the air. This gas plays a major role in atmospheric warming, and changes in its concentration contribute to both natural and human-induced climate change.

When fossil fuels are "burned" to run motor vehicles or machinery, they undergo a combustion reaction, where the fuel reacts with oxygen to release energy. The example below shows the combustion of oil. Oil contains a variety of hydrocarbons. The equation for the combustion of octane, a hydrocarbon with eight carbon atoms, is shown here.



Note the production of carbon dioxide in the reaction. Fossil fuels also contain contaminants such as sulfur and nitrogen that pollute the environment.

Connect to Unit 2,
pages 165 to 167

Fuel Cells

Other forms of energy transformation emit fewer pollutants. In fuel cells, chemical potential energy is transformed into electrical energy when oxygen reacts with hydrogen. If the fuel cell is 100% efficient, the only product is water, as shown below.



Figure 3.20 shows how one type of fuel cell, called a proton exchange membrane fuel cell, works. Hydrogen in tanks and oxygen in air provide a supply of reactants. An electrolyte membrane lets only hydrogen ions pass to the cathode. Electrons from the gas stay on the anode. Hydrogen ions that reach the cathode react with electrons and oxygen to form water. The electrons are drawn from the anode, through an external circuit, to the cathode. This current powers a load.

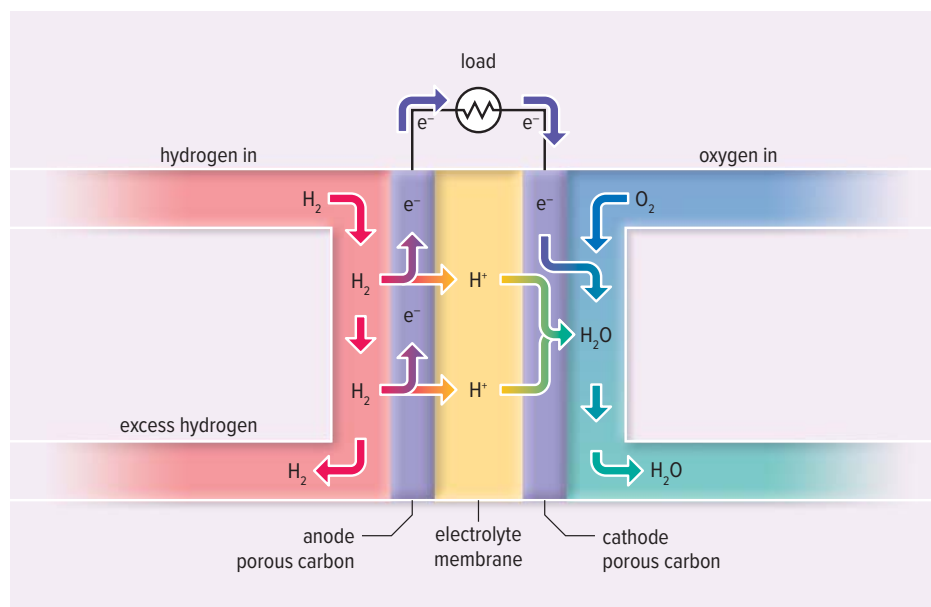


Figure 3.20 The electron flow generated by this proton exchange membrane fuel cell can run a load, such as a vehicle motor.

Extending the Connections

The Future of Fuel Cells

Use the resources available to you (information provided by your teacher, digital or print resources) to find out more about how fuel cells may be used in the future. What is currently restricting the development of future technologies?

Before you leave this page . . .

1. Where is chemical potential energy in molecules stored?
2. How is chemical potential energy transformed by living things?

Energy is transformed in nuclear reactions.

Nuclear reactions are reactions in which changes occur in the nuclei of atoms. In many cases, the atoms are transformed into other elements. As in chemical reactions, energy is transformed in nuclear reactions. However, the amount of energy transformed is much greater.

Types of Nuclear Reactions

Recall that atoms are conserved when energy is transformed in a chemical reaction. The atoms stay the same, but they are rearranged to form new substances. In a nuclear reaction, the atoms of one element are usually changed into atoms of a different element. Often, the atoms involved are isotopes. *Isotopes* are two or more forms of the same element with the same number of protons but a different number of neutrons.

If isotopes have too many neutrons compared to protons, they have too much energy and are, therefore, unstable. Isotopes that are unstable are called *radioactive isotopes*. To become more stable, they get rid of their extra energy by emitting particles from their nuclei. These emissions are often accompanied by radiation. **Radiation** is a mechanism of energy transfer in which atoms or molecules give off energy in the form of electromagnetic waves.

Alpha Decay

Nuclear decay is the change to an atom due to the emission of particles or radiation. The nuclei of some radioactive isotopes decay by emitting alpha (α) particles, which are the same as helium nuclei. This process is called *alpha decay*. When a nucleus emits an α particle, the resulting nucleus has two fewer protons and two fewer neutrons. Therefore, its mass number is reduced by 4 and its atomic number is reduced by 2. Recall that mass number is the number of protons plus neutrons in an atomic nucleus. Atomic number is the number of protons alone. In a nuclear equation, this information is written as shown in **Figure 3.21**, where an α particle is written as a helium nucleus (He). The mass number is written to the left, above the atomic number.

radiation a mechanism of energy transfer in which atoms or molecules give off energy in the form of electromagnetic waves

nuclear decay the change to an atom due to the emission of particles or radiation

Figure 3.21 In a nuclear equation, mass number (upper left) and atomic number (lower left) are written with the chemical symbol, as they are for this helium nucleus.

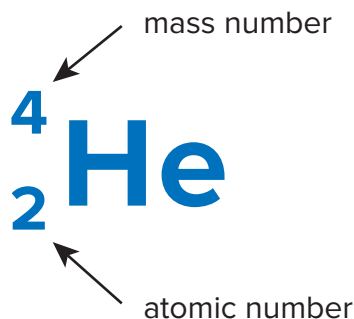
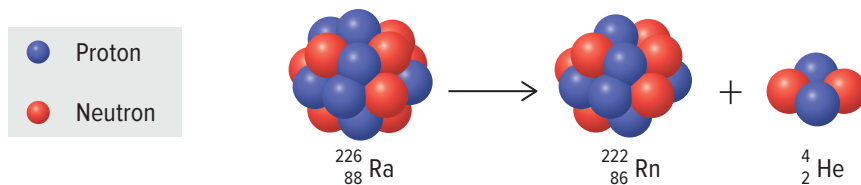


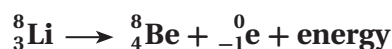
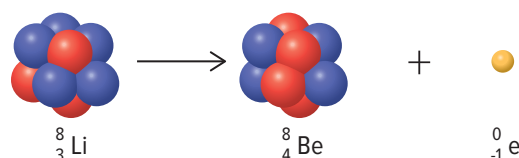
Figure 3.22 shows the alpha decay of the isotope radium-226. Notice that when radium-226 emits an α particle, it becomes a radon-222 atom. The alpha particle is ejected at high speed and has a lot of energy.



$^4_2\text{He} = ^4_2\alpha$, where 4 is the mass number and 2 is the atomic number

Beta Decay

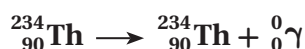
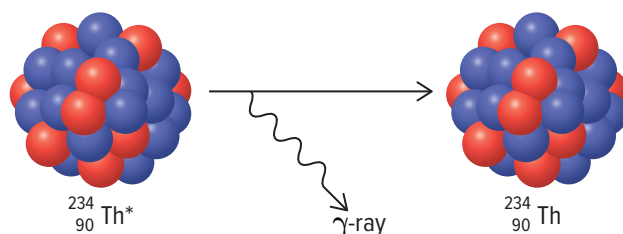
Some radioactive isotopes emit a beta (β) particle, which is a fast-moving electron (e). In this process of beta decay, a neutron in the nucleus decays into a proton and an electron. The electron is emitted as a beta particle; the proton stays in the nucleus. The new nucleus has one less neutron and one more proton. **Figure 3.23** shows the beta decay of lithium-8.



$^0_{-1}e = ^0_{-1}\beta$, where 0 is the mass number and -1 is the atomic number

Gamma Decay

After alpha or beta decay, a nucleus is often left in an excited state, so it is unstable. To become stable, the nucleus gives off a high energy photon called a gamma (γ) ray (**Figure 3.24**). Since a photon is electromagnetic energy, thorium-234 does not become a different atom. The * indicates an excited state.



For $^0_0\gamma$, 0 is the mass number and 0 is the atomic number

Figure 3.22 Alpha decay of radium-226

Figure 3.23 Beta decay of lithium-8 **Analyzing:** What are the products of this nuclear reaction?

Figure 3.24 Gamma decay of thorium-234

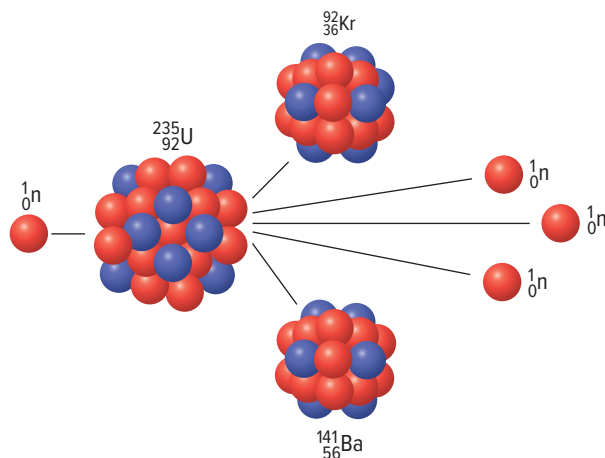
nuclear fission a process in which a heavier nucleus is split into smaller, lighter nuclei with the release of energy

Figure 3.25 When a uranium-235 nucleus absorbs a neutron, it undergoes fission to produce a krypton-92 nucleus, a barium-141 nucleus, and three free neutrons. The krypton and barium isotopes are radioactive.

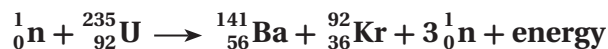
Analyzing: Do krypton-92 and barium-141 have more or less energy than uranium-235? Explain your reasoning.

Nuclear Fission

In **nuclear fission**, a heavier nucleus is split into smaller, lighter nuclei with the release of energy. For example, when uranium-235 is bombarded with neutrons, some of the nuclei absorb a neutron (n). This causes them to



split into two smaller nuclei, releasing energy in the process. The two nuclei can vary among many combinations of elements, depending on how the atom splits. One example is shown in **Figure 3.25**.



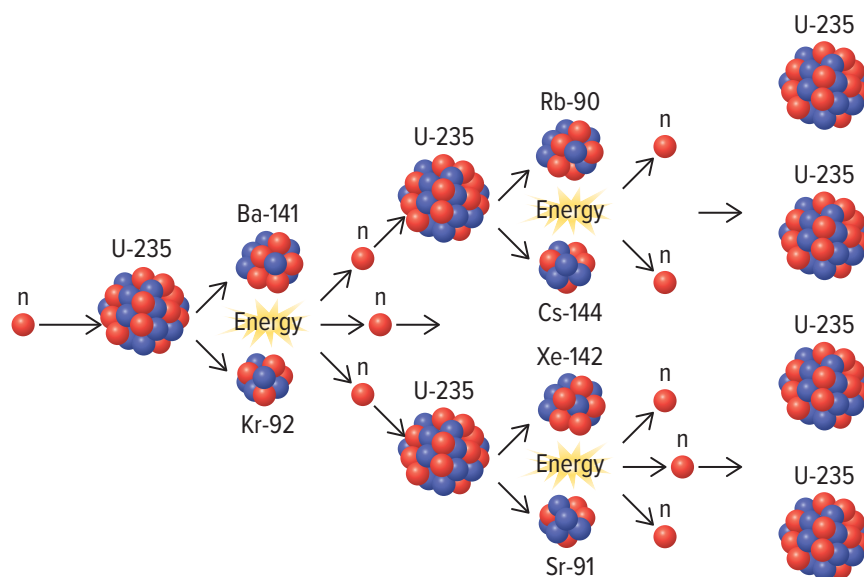
For ${}_0^1\text{n}$, 1 is the mass number and 0 is the atomic number

Fission Chain Reactions

Notice that three neutrons are produced in the reaction in **Figure 3.25**. Each neutron can split a neighbouring uranium-235 nucleus, and each of these produces three more neutrons. In this **chain reaction**, a single reaction causes many further reactions. For nuclear fission, a chain reaction results in the fission of many uranium-235 nuclei. This dramatically increases the amount of energy transformed. An example of a chain reaction is shown in **Figure 3.26**. The products are all radioactive isotopes, so they give off alpha or beta particles and sometimes gamma rays.

chain reaction a process in which one reaction leads to a series of further reactions

Figure 3.26 Two or more neutrons are produced as one uranium-235 nucleus undergoes fission. These neutrons split apart neighbouring uranium nuclei to form other radioactive isotopes and more nuclei. This results in a chain reaction and the fission of more uranium nuclei. **Modelling:** How can you design a model to demonstrate a fission chain reaction?

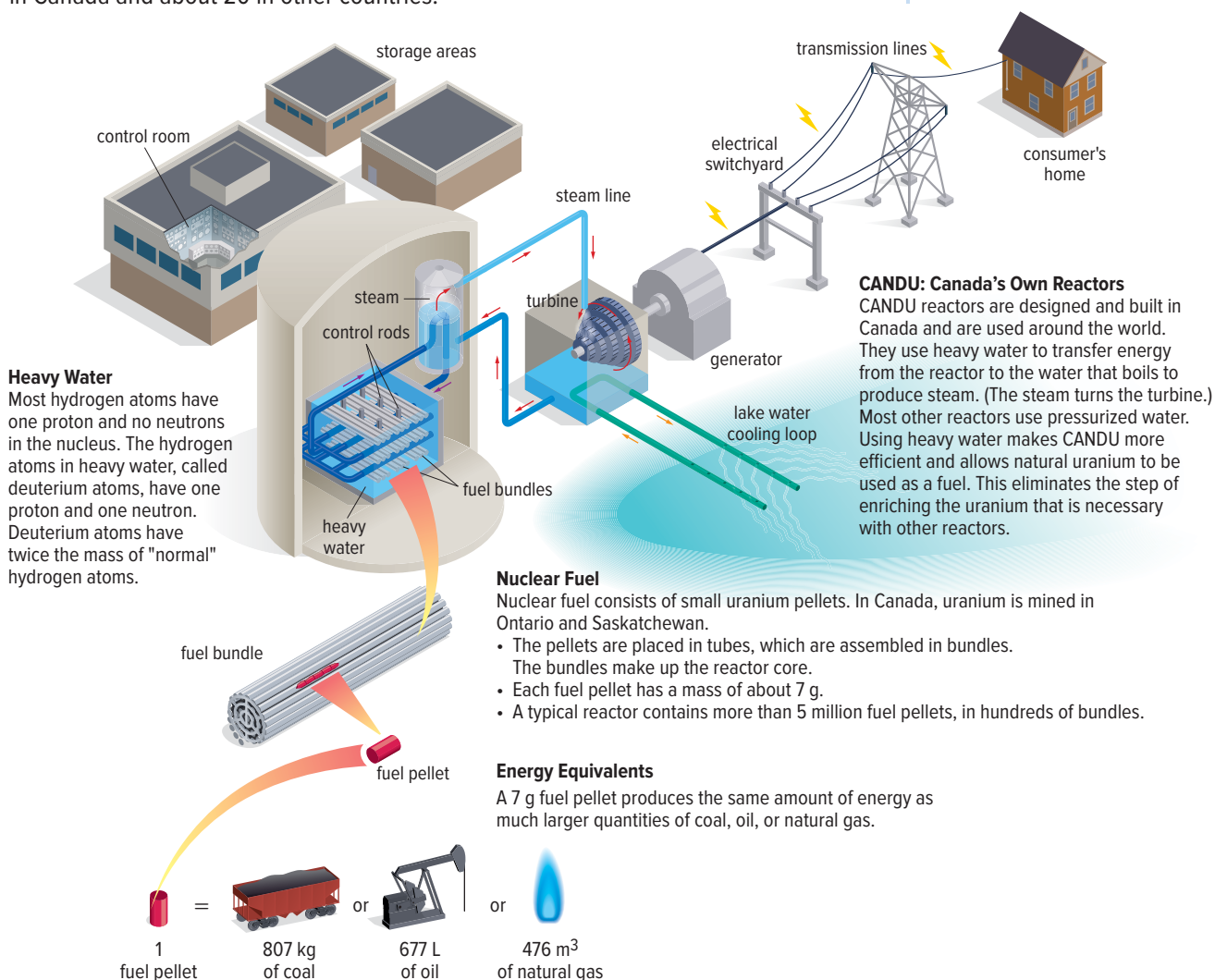


Nuclear Reactors

Nuclear reactors transform the thermal energy produced in nuclear fission reactions into electrical energy. These reactors generally use uranium. Natural uranium consists mostly of the isotope uranium-238. Only about 0.7% is uranium-235, which is the isotope that fuels nuclear reactors. Inside a nuclear reactor, uranium-235 atoms undergo fission. If enough of this uranium is present, the neutrons emitted in the fission of one uranium nucleus can be absorbed by other uranium nuclei. This sets off a chain reaction that releases an enormous amount of energy.

A nuclear reactor contains and controls fission reactions and the energy they release. A liquid coolant flows through pipes close to the fuel and carries thermal energy away from the reactor. The liquid carries the thermal energy to another set of water-filled pipes, which the heat converts into steam. The steam turns turbines that generate electrical energy. The Canadian CANDU reactor in **Figure 3.27** uses natural uranium. This differs from other reactors, which use fuel enriched with uranium-235.

Figure 3.27 At this time, there are 18 CANDU reactors operating in Canada and about 20 in other countries.



nuclear fusion a process in which two very small nuclei combine, or fuse, to form a slightly larger nucleus

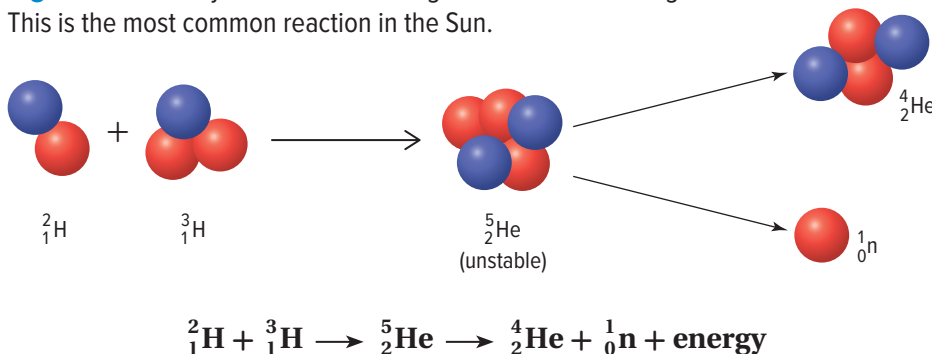
Figure 3.28 Nuclear fusion occurs in all stars.



Nuclear Fusion

In **nuclear fusion**, two very small nuclei combine, or fuse, to form a slightly larger nucleus. For nuclei to fuse, they must collide with a tremendous amount of force. Thus, fusion requires very fast-moving particles to begin. This means it needs to occur at extremely high temperatures. However, once the nuclei fuse, they themselves generate the energy needed for fusion reactions to continue. Scientists have been trying to find a way to control fusion to produce energy, but they haven't been successful. However, fusion reactions occur in the Sun and other stars, like those in **Figure 3.28**. The reaction that occurs most frequently in the Sun is the fusion of two isotopes of hydrogen. These are called deuterium, ${}^2_1\text{H}$, and tritium, ${}^3_1\text{H}$. This fusion reaction forms the very unstable isotope, helium-5. Helium-5 immediately breaks down into helium-4 and a neutron. The reaction is shown in **Figure 3.29**. The helium-4 that is produced is stable. Thus, fusion does not produce radioactive materials.

Figure 3.29 Many small nuclei undergo fusion to form a larger nucleus. This is the most common reaction in the Sun.



Connect to Investigation 3-D on page 240

The amount of energy released in a fusion reaction is about 10 times less than the amount transformed by the fission of uranium-235. However, the sum of the masses of deuterium and tritium is about 47 times smaller than the mass of uranium-235. Therefore, the energy released per unit mass is much larger for nuclear fusion than for nuclear fission. As well, the products produced by fusion are not radioactive isotopes, so no radioactive nuclear waste products result.



Before you leave this page . . .

1. Compare chemical reactions and nuclear reactions in terms of how much energy they transform.
2. a) What is a radioactive isotope?
b) How do these isotopes get rid of their extra energy?
3. How is alpha decay different from beta decay?
4. Why does gamma decay occur?
5. How are nuclear fission and fusion similar? How do they differ?
6. At a glance, it appears like the law of conservation of energy does not apply to nuclear reactions. Explain why it does.

Effects of Radiation on Living Tissues

What's the Issue?

Radiation that is harmful to living things is called *ionizing radiation*. When this radiation interacts with matter, including living tissue, it ionizes atoms. When an atom is ionized, it loses electrons, which often breaks chemical bonds. New bonds may then form, creating different compounds. When this happens in living tissue, molecules, cells, tissues, and even entire organs can no longer perform their normal functions. Alpha, beta, and gamma decay can all ionize atoms.

Alpha particles have about 8000 times the mass and twice the charge of beta particles. Their large size and charge cause alpha particles to deposit all their energy on the surface of the matter with which they collide. If that matter is the skin of living tissue, the energy is often deposited on dead cells, so it does no harm. However, if radioactive isotopes that emit alpha particles are ingested or inhaled, they can damage tissue significantly.

Beta particles can penetrate the surface of matter. If skin is exposed to beta radiation, it penetrates into living tissue and causes serious burns. However, it does not go beyond the skin into other organs unless it is inhaled or ingested.

Gamma radiation has no mass or charge, so it can pass through very dense matter, including lead and concrete, without interacting with it. This means it can ionize atoms and molecules anywhere in the body. When a gamma ray photon does interact with matter, it deposits all of its energy in one interaction, so it is much more dangerous than alpha or beta radiation.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. How do people who work in facilities where they are exposed to ionizing radiation protect themselves?
2. Radiopharmaceuticals are medicines used to treat cancer and other diseases. They contain radioactive isotopes. Radioactive isotopes emit ionizing radiation. How can these harmful isotopes be used to heal the body?

CAUTION



SPECIAL INSTRUCTIONS

CONTACT RADIOLOGICAL CONTROL BEFORE WORKING IN THIS AREA

CONCEPT 3

Energy is transformed when light energy interacts with matter.

absorption a process in which energy is taken up by matter without being reflected or transmitted

When an electron absorbs energy, it jumps to a higher energy level.

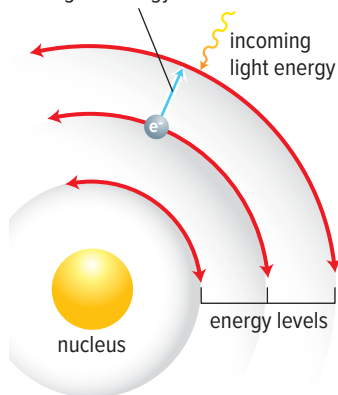


Figure 3.30 When a chlorophyll molecule absorbs light energy, an electron receives this energy and jumps to a higher energy level. As a result, the molecule becomes excited.

Connect to Investigation 3-E on page 241

Chlorophyll in plants absorbs light energy. In **absorption**, energy is taken up by matter without being reflected or transmitted. When light energy is absorbed, an electron in a chlorophyll molecule receives this energy and becomes excited (**Figure 3.30**). In chlorophyll, the energy is used to make high energy bonds. However, in different applications, it is often transformed into other forms of energy. How the energy is used in an excited molecule depends on the properties of the molecule and where it is located. The following examples explore the transformation of light energy in a photovoltaic cell and the human eye.

Photovoltaic Cells

Photovoltaic cells transform light energy directly into electrical kinetic energy (**Figure 3.31**). To make a cell, thin slices of a semiconductor such as silicon are treated with different materials and then sandwiched together. When two types of silicon come into contact in a photovoltaic cell, some of the electrons in one type of silicon jump into the holes in the other. This creates a separation of charge that attracts the remaining electrons to one side and repels them from the other. However, these electrons are held tightly by their atoms. When they absorb light energy, they are freed, move across the cell, and cause a current to flow. The current flows through metal conductors to an electric circuit.

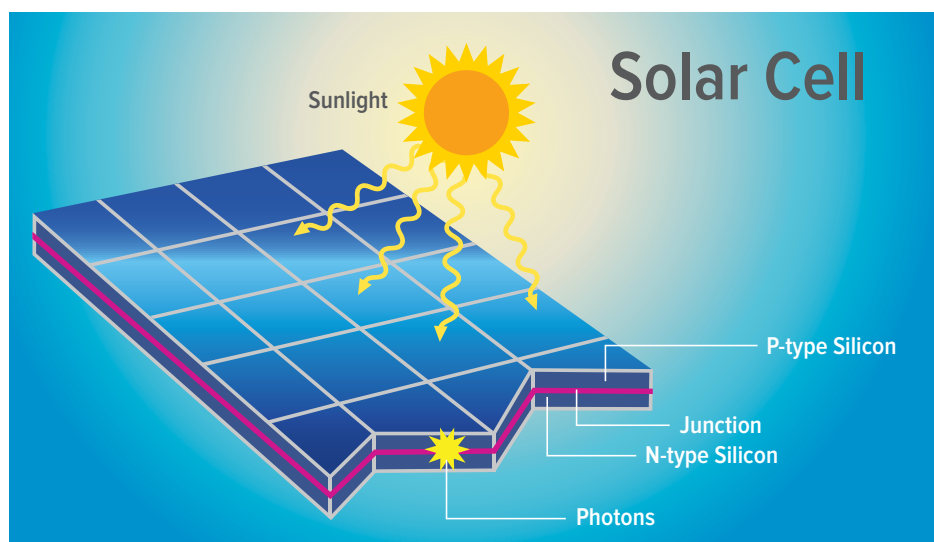


Figure 3.31 Two types of silicon come into contact in a photovoltaic cell. Some electrons in the N-type jump into holes in the P-type. A separation of charge then attracts the other electrons. When these electrons absorb light energy, they are free to move across the cell, and a current flows through conductors into a circuit.

Vision

Vision also involves the transformation of light energy (Figure 3.32). After light energy enters the eye, it is absorbed by rod and cone cells in the retina. These cells contain light-absorbing molecules called rhodopsin. When rhodopsin absorbs light energy, an electron in the molecule becomes excited, causing one part of the rhodopsin molecule to separate from the other. This starts a chain of events that causes a nerve cell to send out an electrical signal. After interacting with other nerve cells in the eye, the signal carries the information to the brain. The brain interprets the information, and you perceive an image.

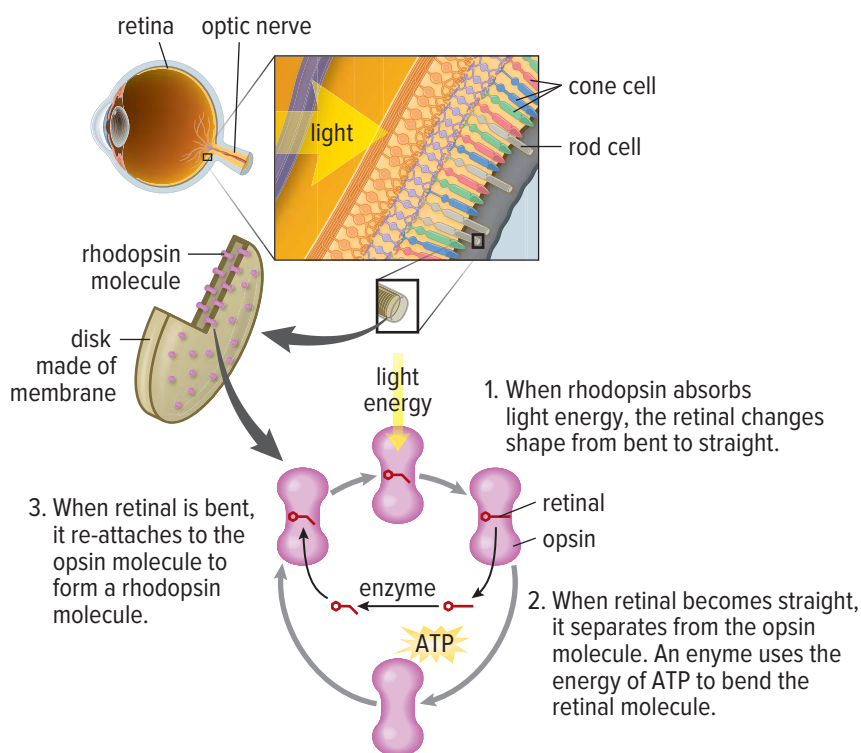


Figure 3.32 Rhodopsin molecules are made up of a protein called opsin and a small molecule called retinal. When the rhodopsin absorbs light energy, an electron in the retinal molecule becomes excited, and the retinal separates from the opsin. This sends a signal to the brain.

Activity

Transforming Light Energy

1. With your group, brainstorm how you could apply the transformation of light energy to achieve a goal or carry out a task of your choice.
2. Create a plan to complete your task and share it with your class. Incorporate any feedback from your class to improve your design.
3. With your teacher's approval, build your design (or create a model if it is not possible to build it).
4. Once you are finished, demonstrate how the task is completed.



Before you leave this page . . .

1. What role do electrons play in transforming light energy?
2. Compare human vision to a photovoltaic cell in terms of energy transformation.

Focus on Physics

Radiation Protection Technician

Biomedical Engineer

Materials Designer

Oceanographer

Environmental Educator

What kinds of jobs are there that involve transformation of energy?



Audiologist

An audiologist needs to understand how sound energy is transformed into mechanical kinetic energy and electrical energy by the ear and nervous system. Hear hear!



Archeologist

Ground-penetrating radar, magnetometers, radiocarbon dating, electrical resistivity—these are some of the tools and techniques related to energy that help archeologists investigate the past.



Meteorologist

A meteorologist can tell you whether there will be a thunderstorm tomorrow, as well as how thermal energy, mechanical kinetic energy, and electrical potential energy are all transformed to produce that storm.

Questions

1. What other jobs and careers do you know or can you think of that involve energy transformation?
2. Research a job or career related to Unit 3 that interests you. What attracts you to it? What kinds of things do you have to know, do, and understand for this job or career?

Make a Difference

Transforming Solar Energy to Make a Difference in Remote Areas

More than 1.2 billion people lack access to electrical energy. Many do not have a reliable source of clean drinking water or access to basic disaster relief supplies. It is especially difficult to supply these resources in poorer, rural areas. However, several Canadian inventors and entrepreneurs are working to meet these needs with some innovative technologies.

The Energy Hub This device uses solar energy to charge portable battery packs for rural communities in sub-Saharan Africa. When the battery packs are drained, they are easily recharged. One Energy Hub can provide electrical energy for up to 100 homes.

The SunSaluter This innovation provides both electrical energy and clean water. It uses Earth's gravity and three bottles of water to move a solar panel. The water drips through a filter that cleans it during the day. As the bottle gets lighter, the panel moves to follow the Sun. The inventor, Eden Full, was a keynote speaker at UBC's Changemaker event, which celebrates making a difference in people's lives.

Solarships These Canadian hybrid aircraft deliver supplies to remote areas. Built to suit their specific missions, their compact design lets them take off and land in the space of a soccer field. They are partially lifted by a buoyant gas and have a large surface area to expose solar panels to maximum sunlight. Solarships can use either solar energy or fossil fuels but emphasis is on the sustainability offered by solar energy.

Apply and Innovate

1. Imagine that you are volunteering with one of these projects. Communicate with a friend back home about your experience.
2. Do you think any of these projects could be useful in Canada? Explain your reasoning.
3. What kind of energy-related action can you take—for example, coming up with an innovative idea, designing a prototype, developing a public information campaign—that can make a difference in the lives of people locally, nationally, or globally?



A solarship being built at the company's assembly and testing facility

Storing Energy in Salt

What's the Issue?

Transformation of solar energy to electrical energy can't occur at night, and it is decreased on cloudy days. This reduces how effectively solar energy can supply electrical energy on a large scale. We can't change when and how much sunlight reaches Earth, but there is a solution to this problem. It relies on salt.

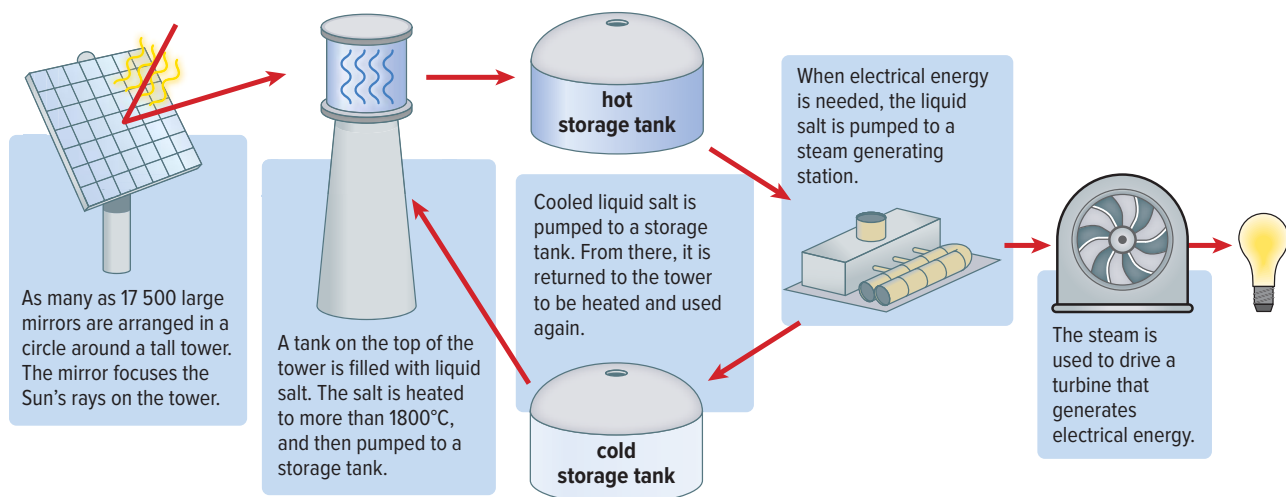
The photo shows an array of thousands of mirrors that are programmed to move with the Sun at Gemasolar in Spain. The mirrors reflect sunlight onto a central molten salt tower. Molten salt is pumped through a coil in the tower. Solar energy is transformed into thermal energy, causing the molten salt to become extremely hot (about 565°C). The salt is then pumped to a power plant, where its thermal energy boils water into steam that drives a turbine to generate electrical energy. Some of the molten salt is also pumped to a storage tank, where it is stored for use at night. This allows the plant to run 24 hours a day.

Dig Deeper



Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. The molten salt used at Gemasolar is not table salt. Use resources available to you, including those provided by your teacher, to find out what kind of salt is used, and why.
2. In the opening pages of this unit, you read that skyscrapers might one day be built of glass that can generate electrical energy. What benefits and problems might be associated with this technology?



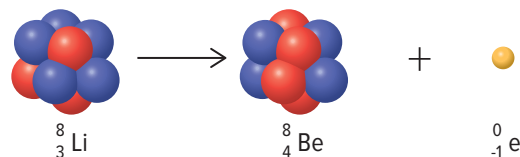
Check Your Understanding of Topic 3.2

OP Questioning and Predicting **PC** Planning and Conducting **PA** Processing and Analyzing **E** Evaluating
AI Applying and Innovating **C** Communicating

Understanding Key Ideas

- What factor determines the amount of energy transformed in a chemical reaction? **PA**
- Indicate whether the following statements are true or false. Correct any statements that are false. **PA**
 - More chemical potential energy is stored in the bonds of some compounds than in others.
 - In all chemical reactions, energy is absorbed from the surroundings when chemical bonds form.
 - Energy released in a chemical reaction is always thermal energy.
 - If the reactants are higher in chemical potential energy than the products, energy is absorbed from the surroundings during the reaction.
 - If the reactants are lower in chemical potential energy than the products, the reaction is endothermic.
- The amount of energy released in cellular respiration is similar to the amount released when methane reacts with oxygen. Why does your body not burn? **PA AI**
- Compare energy transformation in fossil fuel combustion and fuel cells. **PA AI**
 - How are these processes similar?
 - How do they differ?
- What is a radioactive isotope?
 - What role do radioactive isotopes play in nuclear reactions? Use a diagram to support your answer. **PA C**

6. Examine the diagram below. **PA AI**



- What process is occurring here?
- What are the products of the process?
- How does this process differ from a chemical reaction?

Connecting Ideas

- Describe how the processes of cellular respiration and photosynthesis are connected. **PA AI C**
- Explain why a chain reaction is necessary for using nuclear fission to generate electrical energy. **PA AI**
- Describe how the process carried out in a photovoltaic cell is similar to what occurs in human vision. **PA AI C**

Making New Connections

- Nuclear fission generates about 15% of the electrical energy used in Canada. **OP PA E AI C**
 - Describe why nuclear fission is currently used for this purpose rather than nuclear fusion.
 - What are two benefits of using nuclear fusion to generate electrical energy rather than fission?
 - Predict whether fission or fusion will be used to generate electrical energy by the end of the century. Explain your reasoning.
- "We are all nuclear powered." Evaluate whether this statement is true. Justify your response. **PA E AI**

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating

Safety

- Use care in cutting materials.
- Setups and/or materials in this Investigation will generate heat. Work very carefully and conscientiously to protect yourself and your surroundings.

What You Need

- timer
- materials your team's design requires

Solar Cooker Challenge

"It's so hot outside you could fry an egg on the pavement!" You've probably heard this saying before, but is it actually possible for solar energy to cook something? In this Investigation, you will see if you can use solar energy to boil water.

Question

How can you design an apparatus that uses solar energy to boil water? Formulate a hypothesis that you can investigate and evaluate.

Procedure

1. With your group, design an apparatus that solves the problem posed in the question.
2. Brainstorm the materials that you will use, and determine how you will construct your cooker. In your design, consider the following questions.
 - What will you use to reflect sunlight?
 - What shape will direct the most sunlight onto the water?
 - How will you use materials to create the shape?
 - How will you contain the water and keep it in the correct position?
 - How will you orient your cooker to capture the most sunlight?
3. Sketch the design of your cooker. Create a materials list and record any safety precautions. Have your teacher approve these before you continue.
4. Assemble your materials and build your cooker.
5. On a sunny day, attempt to boil water with your cooker. Time how long it takes your cooker to boil the water.

Evaluate

1. How was energy transformed and/or transferred in this investigation?
2. Compare your results as a class.
 - a) Which solar cooker boiled the water in the least amount of time?
 - b) How might you change the design of your cooker to improve its function? If the cooker in part a) was not your own, you might try to apply some of its properties to your own design.

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need

- access to information resources (for example: online, print, interviews)

Fission versus Fusion

You are volunteering at a B.C. company that is trying to find a way to control fusion to produce electrical energy. Part of your job is to create educational materials that compare fission and fusion. These materials are used in presentations to the public, possible funding sources, and other stakeholders.



Fusion Reactor at General Fusion, a Burnaby company involved in fusion research

Question

How can you create and present educational materials that compare fission and fusion?

Procedure

1. With your team, create a summary table that compares fission and fusion. Use the questions below to help.
 - How do the reactions in nuclear fission and fusion differ?
 - What conditions are required for fission and fusion to occur?
 - How are these processes currently being used on Earth?
 - Why is your company choosing fusion over fission? (Consider by-products, energy released, and so forth.)
2. Your teacher will assign you a stakeholder as your audience. Use your table to create your presentation promoting fusion. Be sure to keep your audience in mind when creating the presentation.
3. Give your presentation to a panel of students that represent your stakeholder.

Evaluate

1. Use the feedback provided by the panel to write a paragraph explaining how you could improve your presentation.

Extension

2. Use resources available to you, including those provided by your teacher, to find out more about the current state of fusion research in B.C. and around the world.